

- Nulls

Of the primitive types, integers, floating point numbers, booleans and nulls have a predetermined length specified by their type. Octet strings and UTF-8 strings include a length field that gives their lengths in octets.

A.5. Container Types

The Matter TLV format supports the following container types:

- Structures
- Arrays
- Lists

Each of the container types is a form of element collection that can contain primitive types and/or other container types. The elements appearing immediately within a container type are called its *members*. A container type can contain any number of member elements, including none. Container types can be nested to any depth and in any combination. The end of a container type is denoted by a special element called the ‘end-of-container’ element. Although encoded as a member, conceptually the end-of-container element is not included in the members of the containing type.

A.5.1. Structures

A structure is a collection of member elements that each have a distinct meaning. All member elements within a structure SHALL have a unique tag as compared to the other members of the structure. Member elements without tags (anonymous elements) are not allowed in structures. The encoded ordering of members in a structure may or may not be important depending on the intent of the sender or the expectations of the receiver. For example, in some situations, senders and receivers may agree on a particular ordering of elements to make encoding and decoding easier.

Where a distinguished ordering of members is required (for example, for the purposes of generating a hash or cryptographic signature of the structure), the members of the structure SHALL be encoded as specified in [Section A.2.4, “Canonical Ordering of Tags”](#).

A.5.2. Arrays

An array is an ordered collection of member elements that either do not have distinct meanings, or whose meanings are implied by their encoded positions in the array. An array can contain any type of element, including other arrays. All member elements of an array SHALL be anonymous elements – that is, they SHALL be encoded with an anonymous tag.

A.5.3. Lists

A list is an ordered collection of member elements, each of which may be encoded with a tag. The meanings of member elements in a list are denoted by their position within the list in conjunction with any associated tag value they may have.

A list can contain any type of element, including other lists. The members of a list may be encoded